

KANNAN DHARMALINGAM

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Passionate & inspirational leader promising innovation, promoting growth and profitability, and evolution on every layer or level within a company; mainly Start-up, MNC, etc.

Hands-on technology leader with 15+ years of experience, specializing in UI and cloud-native architectures, scalable data pipelines, and distributed systems. Proven track record in leading end-to-end solution delivery across high-volume analytics platforms, enterprise-grade applications, and modern web technologies. Expert in frameworks such as React, Angular, Node.js, and Python, with strong experience in Azure cloud services, Kubernetes orchestration, and data-driven workflows. Passionate about leveraging cutting-edge tools like Zustand, React Query, Tailwind CSS, and component libraries to build maintainable, high-performance user interfaces.

Key Strengths involved: Architecture Design & Report Generation | New Feature Implementation | Performance & Test Automation | Stakeholder Management Continuous Integration & Deployment | Team Management & Leadership | Effective multi-tasker and team builder | Design Patterns - MVVM, MVP, MVC | Agile Methodologies | Cloud and Microservices | Product or Program Management | Strategic Planning & Direction | Engineering Operations | Product Designing & Implementation | Digital Strategy | Process Enhancement | Innovation, Consulting, and Change.

Certifications:

AWS [Solution Architect Associate](#) | AWS [Certified DevOps Engineer - Professional](#) | AlgoExpert [System Design](#) and Solution Architecture.

BENCHMARKS PERFORMANCE

- **Architected and developed reusable React components and front-end libraries**, optimising performance, scalability, and maintainability of complex web applications. This reduced development time and significantly improved code reuse.
- **Led a 35-member development team at Sumeru**, elevating team efficiency by 40% across high-profile client projects. Delivered secure, high-quality products consistently, driving operational excellence and positioning the team as a key asset in meeting critical business objectives.
- **Spearheaded the migration of a legacy Angular 1.5 system to Angular 14**, ensuring a seamless transition and modernization of the tech stack. This migration significantly improved application performance, security, and maintainability.
- **Migrated front-end technologies to React**, increasing overall application speed and scalability, while integrating with modern backend services like Node.js and cloud infrastructure (AWS), achieving a 30% performance boost.
- **Pioneered the conversion of legacy applications into a SaaS model using React and Node.js**, driving new revenue streams and improving application scalability to support multi-tenant systems.
- **Served as the Single Point of Contact (SPOC) for front-end development, particularly in React and mobile applications**, delivering cohesive strategies for high-quality product development and deployment in the financial sector.
- **Architected and implemented React Native-based iOS apps**, enhancing product performance and reducing app load times by 20%, while leading test automation to ensure robust performance under varied conditions.
- **Designed and architected full-stack products for ENSOURCE and ENCreditable with React, Node.js, and MongoDB**, delivering scalable, secure, and high-performing solutions aligned with market demands.
- **Led the migration from Angular 1.5 to Angular 14**, improving front-end architecture, boosting performance, and enabling better maintainability for large-scale enterprise applications.
- **Developed an internal dynamic form builder using Angular**, improving operational efficiency for loan application processing by reducing form handling complexities—this is maintained for legacy support.
- **Directed the full-stack development of Facctum Transaction Screening and Alerts Management with React**, ensuring a high-performing, scalable SaaS architecture to meet client demands.
- **Received numerous performance awards at Sumeru** for leading complex projects, driving architectural excellence, and delivering high-quality, innovative solutions across industries.

CAREER DETAILS

Catalys as Chief Technology Officer | Since Oct 2024

Driving the technology architecture, roadmap, and data engineering for a performance marketing platform, focusing on scalable data pipelines, cloud-native deployments, and modern frontend frameworks to support high-volume analytics and marketing operations.

- **Architected and delivered a cloud-native data platform leveraging Python and Azure services** to support heavy data ingestion, transformation, and reporting pipelines, enabling real-time campaign insights at scale.
- Led container orchestration using **Azure Kubernetes Service (AKS)**, improving deployment workflows, fault tolerance, and system scalability across distributed environments.
- **Modernized frontend architecture by integrating React, Zustand for state management, React Query** for optimized data fetching, and Tailwind CSS for efficient, responsive UI development.
- Standardized UI component development using **Shadcn's** library, improving consistency, accessibility, and team productivity across multiple features.
- **Designed and implemented robust API architectures** and data workflows, ensuring low-latency access to analytics data while addressing security, compliance, and scalability requirements.
- Established observability and monitoring strategies integrating metrics tracking and automated deployment pipelines, improving system reliability and developer velocity.
- Collaborated with cross-functional teams to align technology initiatives with marketing analytics goals, driving feature prioritization, data governance, and scalable architecture planning.
- **Mentored engineering teams on best practices for cloud deployments**, state management, and CI/CD pipelines, fostering a culture of performance, scalability, and innovation.

Facctum as Lead Consultant(Solutions Architect) | Since Sep 2024

Created strategic and annual implementation plans with the team, defining key outputs and outcomes, while maintaining smooth partner relationships to drive successful product development.

- **Architected and developed a highly scalable transaction monitoring and alert management application for a bank's AML division**, integrating a GraphQL API to enable efficient querying and reporting of transaction alerts. This enhanced the bank's ability to detect and respond to suspicious activities, ensuring compliance with regulatory requirements.
- **Established IT service delivery and governance frameworks**, considering business objectives, compliance mandates, and security requirements, ensuring the successful alignment of technology with business goals.
- **Championed digital transformation programs**, rolling out initiatives like IoT, mobility, and enterprise API platforms, and led the development of an analytics roadmap integrating both standard BI and ML-based analytics to drive insights.
- **Set a benchmark within ATC by developing and implementing information architecture programs**, including master data management platforms, data governance, and data quality assurance processes.
- **Set up key performance indicator (KPI) metrics** for reporting team members, enhancing accountability, performance tracking, and driving continuous improvement within the team.
- **Led decision-making on enterprise architecture** and detailed component-level software design, overseeing the timely build and delivery of scalable software platforms.
- **Built complex, consumer-facing digital products and internal systems**, across web and mobile applications in high-pressure environments, ensuring robust performance and scalability.

Sumeru Enterprise Tiger Business Solutions as Deputy CTO | Jan 2016-Jan 2023

Led the technical architecture for the full stack development team, managing both front-end and iOS mobile aspects using React Native, while leveraging Node.js for backend development. Architected dynamic reporting solutions and coordinated performance testing to ensure optimal functionality and timely delivery.

- **Designed, researched, and executed strategic projects** included in the Accelerator partnership and project-based support initiatives, driving innovation and business impact.
- **Led the team to deliver continuous product and engineering improvements every sprint**, adopting a productive and iterative approach to ensure timely releases and high-quality standards..
- **Enabled decision-making on enterprise architecture and component-level software design**, overseeing the full lifecycle from architecture through to build and final delivery of scalable software platforms.
- **Advised strategic direction on technology usage** based on deep understanding of customer requirements and business objectives, ensuring alignment with goals.
- **Analyzed data and developed actionable recommendations**, based on in-depth research findings to enhance product and business performance.
- **Managed overall service delivery and SLA management**, monitoring performance against key success metrics and proactively identifying and mitigating risks.
- **Determined appropriate staffing levels and skill sets**, ensuring project objectives and tasks were completed within budget and resource parameters.
- **Enhanced core product features with cutting-edge front-end design**, serving as the Single Point of Contact (SPOC) for all new feature development, and ensuring seamless execution.
- **Prepared comprehensive reports for all 3 products**, utilizing FusionCharts, D3.js for visual data representation, and Elasticsearch-backed Python reports for detailed insights.

- **Collaborated closely with the product head** to implement new features and coordinated their allocation to team members for timely delivery. Worked with performance testing and product automation teams to ensure product robustness.
- **Implemented web-security best practices**, including vulnerability assessments and penetration testing, ensuring compliance with industry security standards.
- **Deployed the Diffie-Hellman algorithm for client authentication and data encryption**, enhancing the security framework across products.
- **Developed a GraphQL API** to streamline data querying and improve client-side performance by enabling efficient, flexible data retrieval from multiple sources.
- **Executed a POC using AWS Lambda** for serverless computing, demonstrating reduced infrastructure costs and enhanced scalability for application features.
- **Deployed and managed containerized applications using Amazon EKS**, optimizing scalability and operational efficiency across microservices architecture.
- **Developed and executed robust development, documentation, and product engineering processes**, ensuring streamlined operations and efficient project management.
- **Ensured prompt resolution of client issues**, maintaining professional communication and addressing concerns in a timely manner while effectively communicating the brand message.

Tesark Technologies as Technical Lead (Joined as Associate Software Engineer) | Aug 2013-Dec 2015

Gathered and analyzed client requirements, collaborating closely with the development team for smooth implementation and timely delivery of projects.

- **Assisted in training new employees**, ensuring their integration into the team while mentoring them in technical and project best practices.
- **Reviewed team performance** and worked with the software development team to address technical debt and improve code quality.
- **Prepared and presented progress reports**, keeping stakeholders informed on project briefs, requirements, and deliverables.

SEO Media website promotions as Independent Consultant - Static website designing & development | Jun 2010-Jul 2013

Designed and developed static websites with a focus on user experience and SEO optimization to improve search rankings.

- **Advised clients on media promotion strategies and implemented SEO best practices** to enhance online visibility.
- Created **web style guides** to ensure design consistency across client projects.

KEY PROJECTS/PRODUCTS

- **FacctShield**: Architected and developed a highly scalable transaction monitoring and alert management application, enhancing the bank's ability to detect and respond to suspicious activities within their AML division
- **ENCollect**: Led the development of ENCollect, an application designed to manage all communication activities for all stages of collections.
- **ENSource**: Spearheaded the creation of ENSource, offering customer onboarding automation and digitization tailored for banks, financial institutions, and telecom companies.
- **ENCredible**: Pioneered the development of ENCredible, a Digital Collection-as-a-Service (DCaaS) platform empowering banks and financial institutions to expedite consumer analysis and treatment plan implementation.
- **CEGHomes**: Architected and developed CEGHomes, a hostel counseling application streamlining room allocation and payment collection processes. Notably introduced online payment for hostel fees, catering to over 4500 attendees across three days.
- **LAPAROFIT**: Engineered LAPAROFIT, an application facilitating live hospital surgery conferences.
- **1Mark**: Developed 1Mark, an online MCQs test portal catering to school students.
- **The Little Shopper**: Designed and developed The Little Shopper, an online ecommerce application tailored for clothes shopping.
- **Liquidzz**: Orchestrated the creation of Liquidzz, a POS application tailored for juice bars and restaurants.

SYSTEM DESIGN KNOWLEDGE

Microservices Architecture:

- **Led the development of scalable applications (ENSource, FacctShield, ENcredible, Performity)** using Node.js and a microservices architecture to improve performance and maintainability.
- **Implemented horizontal scaling**, including load balancing and sharding, to manage high user loads and ensure responsiveness.

Caching Strategies:

- **Managed application performance** by implementing caching strategies such as cache-aside, read-through, and write-through to optimize system responsiveness and reduce load times.
- **Familiarity with tools like Memcached, Redis**, and other caching solutions to enhance application performance. Implemented a cache-aside strategy with Redis to reduce database load for frequently accessed product data in ENSource and FacctShield.

Data Management:

- **Experienced with NoSQL databases** (e.g., MongoDB, CosmosDB) for scalable data storage and querying in growing applications.

Monitoring and Observability:

- **Proficient in Grafana** for data visualization, Prometheus for metrics monitoring, and Fluentd for log forwarding to gain comprehensive insights into application health and performance.

Security:

- **Applied web security best practices** to design robust and secure systems.

Message Queues:

- **Familiarity with message queues** like Kafka (AWS MSK) and RabbitMQ for asynchronous communication between microservices in Transaction Screening and Monitoring applications.
- This enables efficient processing of high volumes of data and ensures decoupled communication within the architecture.

Cloud Platforms & Infrastructure:

- **Knowledge of cloud platforms (AWS)** and containerization technologies (Docker, Kubernetes) for designing cloud-deployable systems.
- **Extensive experience with Kubernetes and Docker** within AWS environments.
- **Proficient in event-based architecture**, API Gateway, Lambda, Amazon EKS, CloudWatch, Argo Events, and other related tools for building scalable and resilient cloud-native applications.

Notification and Log Storage:

- **Experience with NAT.io** for efficient notification and log storage in the Transaction Screening and Monitoring application.

Scalable College Management System:

- **Technology Stack:** Ruby on Rails, PostgreSQL
- **Key Features:** Automated hostel counselling, managing over 5,000 admissions within three days.
- **Scalability:** Built to handle high volumes of transactions and user requests.

Live Streaming Application:

- **Technology Stack:** Ruby on Rails, WebRTC
- **Key Features:** Real-time streaming for medical conferences.
- **System Design:** Focused on low latency and high availability to ensure smooth live streaming.

Banking Product Suite:

- **Technology Stack:** Angular, React, Node.js
- **Key Features:** Dynamic report generation, iOS app development.
- **System Design:** Modular architecture for easy integration and scalability.

Dynamic Form Handling Library:

- **Technology Stack:** Angular, Node.js
- **Key Features:** Enables quick deployment of new loan products.
- **System Design:** Flexible and reusable components to handle varied loan application fields.

AML System with Microservices:

- **Technology Stack:** Node.js, React, Docker, Kubernetes
- **Key Features:** Transaction screening, alert management, case manager solution.
- **System Design:** Microservice architecture for independent deployability and scalability.

AWS Cost Optimization:

- **Implemented cost-saving measures** through AWS reserved instances and auto-scaling to optimize infrastructure expenses.
- **Leveraged AWS Lambda and serverless architecture** to reduce operational costs while maintaining high availability.

High Scalability and SaaS:

- **Architected high-turnover SaaS applications** with a focus on scalability and resilience.
- **ENCollect supports over 50,000 users** with efficient debt collection management.
- **Transaction Screening handles over 300,000 transactions** daily, ensuring robust compliance and monitoring.

TOOLS AND TECHNOLOGIES

- **Server-Side Programming & Web-application Framework:** Ruby-On-Rails | Express & NodeJS | Python & FastAPI
- **Client-Side Scripting Languages & Framework:** ECMAScript 6 to latest | JavaScript - Underscore & Lodash | jQuery | TypeScript (React & Angular) | **D3.js** | Aurelia | Fusion Charts | Progressive Web Application | JSON | AJAX | Twilio | **GraphQL** | Ngx-

Formly | Angular | Google Maps | Micro-Front-End

- **Frontend Libraries and Frameworks: Angular:** Angular: AngularJS 1.x | Angular 2-14 | Angular Formly | Angular Material | Angular Universal. **React:** ReactJS | React & Redux | React Native | Redux-Saga | Redux Toolkit | Webpack | Flux | MobX | React Hooks.
- **Unit Testing:** Mocha & Jest | Jasmine and Karma
- **Messaging & Communication:** AWS SQS | Kafka | RabbitMQ
- **Database Management System:** MySQL | MongoDB | PostgreSQL | Redis
- **Application Performance & Testing Tools:** JMeter | Octoperf | Telemetry .
- **Design Languages & Web Platforms:** HTML | CSS | SASS | LESS | SCSS | Bootstrap | Foundation | MUI (Material-UI).
- **Cloud Platforms & Infrastructure:** AWS | Digital Ocean | Kubernetes | Docker | API Gateway (Traefik) | Load Balancer (AWS ELB, ALB) | Traefik
- **Application Performance Management (APM) & Monitoring Tools:** Grafana | Prometheus | ELK Stack (Elasticsearch, Logstash, Kibana)
- **Server & Version-control systems:** Git | Bitbucket | TFS | JIRA | Jenkins | CI/CD | SRE Practices.
- **Content Management Systems:** WordPress | Joomla | Blogger
- **Development Tools:** NPM | Bower | Grunt | Gulp | Webpack | Xcode | Android Studio.

EDUCATION

- **Bachelor of Engineering-Computer Science** from Arignar Anna Institute of Science and Technology in 2010

Date of Birth: 05/07/1989 | Languages Known: English, Tamil, Telugu and Kannada